



Problem-Solution Fit

Date	30 June 2026
Team ID	XXXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed **Credit Card Approval Prediction** solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

This matches the original format, with only the topic changed to your project: **Credit Card Approval Prediction**.

1. CUSTOMER SEGMENT(S) [CS] • Banks and Financial Institutions • Credit Card Departments • Loan Officers / Credit Analysts • Credit Card Applicants (Individuals)	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] • Quality and availability of data • Budget for advanced systems • Need for secure and reliable infrastructure • Integration with existing systems	5. AVAILABLE SOLUTIONS PROS & CONS [AS] • Manual Evaluation – Accurate but slow and error-prone • Rule-Based Systems – Fast but inflexible • Traditional Scoring Models – Simple but less accurate • Machine Learning Models – Accurate and adaptive but need quality data and maintenance
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] • Manual credit card approval is time-consuming and costly. • Risk of human errors and inconsistent decisions. • Difficulty in evaluating large number of applications. • High rate of fraud and default risk. • Occurs daily with thousands of applications.	9. PROBLEM ROOT / CAUSE [RC] • Manual verification and subjective judgment. • Lack of standardized evaluation process. • Increasing volume of applications. • Incomplete or scattered applicant information. • Inability to identify high-risk applicants early.	7. BEHAVIOR + ITS INTENSITY [BE] • Applicants expect quick approval and transparency. • Banks need to minimize risk while maximizing customer acquisition. • High intensity – decisions impact revenue and customer trust. • Frequent behavior – applications received daily in large volume.
3. TRIGGERS TO ACT [TR] • Increase in credit card applications. • Need to reduce approval time. • Rising customer expectations. • Higher default and fraud rates. • Need for data-driven and accurate decision making.	10. YOUR SOLUTION [SL] • Credit Card Approval Prediction System using Machine Learning. • Predicts whether an application will be Approved or Rejected. • Built with Flask, Python, Scikit-learn, Pandas, NumPy and Joblib. • Provides fast, accurate, and consistent predictions. • Helps banks reduce risk, save time, and improve customer satisfaction.	8. CHANNELS OF BEHAVIOR [CH] ONLINE • Bank Websites • Mobile Banking Apps • Online Credit Card Application Portals • Customer Dashboards OFFLINE • Bank Branches • Customer Service Centers • Loan Officer Desktops
4. EMOTIONS BEFORE / AFTER [EM] BEFORE: Anxiety, uncertainty, long waiting time, frustration due to unclear decision. AFTER: Relief, satisfaction, trust in the process, confidence in faster approval.		OFFLINE • Printed Forms / Documents • Branch Consultations • Physical Verification